

Reliable and explainable seismic hazard forecasting for underground mine monitoring

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Research code: <https://github.com/junjie-yang11/seismic-bump-forecasting>

Abstract

A common-cohort evaluation links mine-monitoring signals to explainable seismic warning decisions and quantifies the inspection workload those decisions create. Using 2,578 UCI Seismic Bumps mirror records, we assess logistic regression (LR), bagged CART and training-tuned XGBoost on the same 2,063 record-order test rows. Matching the test rows reduces the apparent random-versus-record-order average-precision gap by 89.7 percent for LR and 86.9 percent for CART, demonstrating the importance of evaluation-cohort composition. Training-prior correction improves LR expected calibration error from 0.2904 to 0.0271 and Brier score from 0.1462 to 0.0439. Nine engineering feature combinations assess the information in hazard ratings, geophone signals, seismic activity and shift type. Exact TreeSHAP identifies total seismic energy, low-energy event counts and pulse activity as recurring contributors to XGBoost predictions; importance-rank correlations across four test phases range from 0.712 to 0.941, with changing top-five membership. At training reference budgets of 10 and 20 percent, supplementary holdout XGBoost produces 30 and 81 alerts and detects 3 and 6 of 26 hazardous shifts. The resulting framework connects comparable validation, interpretable monitoring signals and measurable detection-workload tradeoffs. Record order supplies the forecasting proxy, and paired intervals quantify uncertainty conditional on fixed predictions.

Keywords: mining engineering; seismic hazard forecasting; engineering feature ablation; TreeSHAP; explanation stability; warning budget

1 Introduction

Underground mine monitoring combines measurements of rock-mass activity with assessments of dangerous conditions. Forecasts become useful for engineering review when measured signals can be traced through a model to a warning decision and its inspection demands. This requires evidence on the information supplied by each monitoring source, the consistency of its use across operating phases, and the hazardous shifts detected at a defined review workload.

The UCI Seismic Bumps data provide a defined next-shift forecasting task: each record summarizes an eight-hour shift, and the target indicates whether the following shift contains a seismic event above 10,000 J [1]. Existing seismic and seismoacoustic hazard ratings appear alongside energy, pulse and event-count measurements. Their coexistence makes the additional predictive information in measured signals testable against the recorded assessments. The target is next-shift high-energy seismic occurrence; confirmed rockburst accidents are outside its label definition.

Ranking metrics describe how well a model orders hazardous shifts, while engineering evaluation also needs to explain which signals shape that ordering and what a warning rule demands. Validation protocols can change both the training history and the test population [3,4], making a common test cohort essential for interpreting their difference. Model explanations reveal how monitoring inputs contribute to fitted predictions, and frozen thresholds connect those predictions to detection counts and inspection workload.

This study integrates these assessments in one reproducible framework. A matched cohort establishes the comparison across three model families; nine prespecified feature combinations separate recorded hazard assessments from geophone, seismic-activity and operational information. Training-prior correction evaluates the probability scale, and exact TreeSHAP traces feature contributions across four test phases. Four fixed training reference budgets then express warning policies as detections, false alerts and missed hazardous shifts. The framework provides an engineering account of what monitoring information contributes, how a predictor uses it and what its warnings require.

2 Data and engineering feature hypotheses

2.1 Dataset and provenance

The original UCI dataset contains 2,584 records from two longwalls of a Polish coal mine, with 170 positive target records [1]. The positive count refers to labelled shift records, rather than a catalogue of 170 independently identified seismic events. The CSV mirror retains 2,578 records [2]. A direct ARFF-to-CSV audit confirms first-occurrence deduplication without reordering: six excluded original rows, 90, 91, 973, 974, 1018 and 1019, are exact duplicate negatives. All positive records remain. The retained original row IDs are 88, 89, 971, 972, 1016 and 1017, respectively.

Row order is used as the temporal proxy because explicit timestamps and longwall identifiers are absent. The initial training block contains 15.92 percent positive records, compared with 6.98, 1.94, 4.65, 3.49 percent in the four subsequent test phases. Exact phase counts and prevalence are provided in Table 6. The full dataset prevalence is 6.59 percent, whereas the common test cohort has 88 positives among 2,063 records, or 4.27 percent. These differences motivate both matched evaluation and phase-specific analysis.

2.2 Engineering feature groups

Table 1 partitions the existing 17-column design into four engineering groups. The three hazard-rating variables encode recorded assessments, while the geophone and seismic-activity groups contain monitoring measurements. Shift type describes preparation or coal-getting activity. These groups define comparisons between existing hazard assessments and the measured signals available to a forecasting model.

Table 1. Engineering feature groups and the predictive hypotheses examined.

Group	Variables	Question
Hazard ratings	seismic, seismoacoustic, ghazard	How much information is already represented by recorded assessments?
Geophone	genergy, gpuls, gdenergy, gdpuls	Do energy, pulse activity and deviations add information beyond ratings?
Seismic activity	nbumps2 to nbumps89, energy, maxenergy	Do event counts by energy band and released-energy measures add information?
Operation	shift	Does shift type provide predictive context?

Categorical fields retain the existing ordinal encoding. The total count nbumps is excluded because it is the sum of the energy-band counts; constant energy-band columns remain in the audited design. LR uses constant-safe training standardization and L2 regularization. The design uses the supplied shift summaries; constructing cross-shift lag features would require known longwall continuity.

3 Methods

3.1 Common test cohort and model comparison

Record-order evaluation divides the mirror into five consecutive blocks. Four expanding-window fits train on earlier blocks and predict the next block; the first block supplies training only. Every model and feature combination therefore evaluates the same 2,063 rows. A supplementary holdout trains on the first 70 percent and evaluates the last 774 records, containing 26 positive targets. This holdout was inspected in the earlier project, so it is reported as descriptive supplementary evidence and is not used to select a feature set, model or budget.

LR retains L2 penalty 1.0, an unpenalized intercept and balanced class weights. Bagged CART retains 60 ordinary bootstrap trees, maximum depth 6, minimum leaf size 20, unweighted Gini and forest seed 7. XGBoost uses unweighted binary logistic loss and histogram trees, learning rate 0.05, minimum child weight 10 and L2 penalty 5.0, with full row and column sampling, seed 7 and two CPU threads [6]. Its fixed candidate grid combines depth 2 or 3 with 80 or 160 boosting rounds.

Each XGBoost search uses only its supplied training rows: the final 20 percent forms an inner reference, and candidates are fitted on the preceding 80 percent. Selection maximizes tied-score average precision (AP), with negative Brier score as the predefined fallback for a zero-positive reference; the first grid entry wins exact ties. The selected model is refitted on the entire supplied training set. Threshold-reference models conduct their own search within the earlier inner-fit prefix. For feature ablations, full-feature training selects parameters once per fold and the same choices are used for all nine feature combinations.

The random protocol uses ten stratified folds and five seeds, 0 to 4. XGBoost selects parameters through a stratified inner split within each outer training fold. Random predictions are restricted to the record-order test cohort before calculating comparative metrics. Reported random AP is the mean of five seed-specific AP values, rather than AP after averaging scores. LR and CART retain fixed settings; XGBoost is training-tuned. Comparisons characterize these stated workflows rather than equally extensive searches for every model.

3.2 Feature comparisons and conditional uncertainty

The prespecified feature sets are the full design, each group alone and the design with each group removed. They are all reported, without choosing a final feature set from test outcomes. The full-versus-ratings-only contrast addresses the additional predictive information in monitoring measurements and shift type. Group-removal contrasts assess predictive dependence on information groups in the presence of correlated alternatives.

AP is the main ranking measure and groups tied scores before integration. ROC-AUC, Brier score and ten-quantile-bin expected calibration error (ECE) provide complementary discrimination and probability assessments. LR probability assessment uses each outer training fold's unweighted prior to correct its balanced-weight output; CART and XGBoost are assessed as fitted. No test-label prior is used in forecasting or calibration.

Paired moving-block intervals use 2,000 replicates, block length 32 and RNG seed 20261002. Contiguous blocks are sampled separately within each test phase, preserving phase sizes and applying identical sampled rows to every compared score vector. Intervals are the 2.5th and 97.5th percentiles of paired AP differences and condition on fixed predictions. Random-protocol differences average the five seed-specific AP values within each replicate. The earlier LR/CART protocol analysis also checks lengths 16 and 64. The intervals are descriptive, unadjusted for multiple feature contrasts, and exclude model-refit and parameter-selection uncertainty [5].

3.3 Explanations and phase stability

Full-feature XGBoost models explain their own outer-test predictions through native exact TreeSHAP, using the training-derived leaf covers as the tree-path-dependent reference [7,8]. Each row has 17 feature contributions and a bias term. Their sum reconstructs the raw log-odds margin; applying the logistic function reconstructs the predicted probability. Contributions are additive on the model's log-odds scale and describe fitted predictive associations.

Global importance is mean absolute contribution on the common test cohort. Each phase also has its own importance ranks. Pairwise Spearman correlations compare the ranks of all 17 features, with average ranks for ties; top-five Jaccard overlap measures agreement in leading features. A fixed feature-name order resolves ties at the top-five boundary. Value-contribution Spearman associations and scatter plots describe how observed values relate to contributions. Together, the rank and overlap measures characterize recurring feature use and changes among leading inputs.

Illustrative cases use the earliest record in each true-positive, false-positive and false-negative category under the predeclared 10 percent training reference budget. Case selection is a post-evaluation explanation step, independent of model and threshold selection.

3.4 Training reference budgets and warning outcomes

The reference budgets are fixed at 1, 5, 10 and 20 percent. For each outer fold, an inner model predicts the last 20 percent of the training history. The threshold excludes the boundary score and all tied scores together, so at most $\text{floor}(\text{reference size} \times \text{budget})$ reference records are flagged. After outer refitting, the numerical threshold is frozen and evaluated on future test records. All budgets use the same model scores; no test labels enter threshold selection.

Reference budgets are experimental workload scenarios, not established operating limits for a mine. The future actual alert rate is measured separately because refitting and changing score distributions can alter it. Warning outcomes include true detections, false alerts, missed hazardous shifts, precision and recall. The target and available labels measure shift-level detections, not event counts, spatial warning coverage or avoided accidents.

4 Predictive contribution of monitoring information

4.1 Cohort sensitivity and matched model comparison

Table 2 compares validation protocols on identical test records. Full-feature XGBoost record-order AP is 0.0862, compared with 0.0838 for LR and 0.0784 for CART. The XGBoost-minus-LR paired difference is 0.0024 with interval $[-0.0290, 0.0291]$, and the XGBoost-minus-CART difference is 0.0079 with interval $[-0.0019, 0.0267]$. These paired comparisons establish the model context for the monitoring-information and warning-policy analyses below.

Table 2. AP on identical 2,063 test rows and paired random-minus-record-order intervals; random values average five seeds.

Model	Random AP	Record-order AP	Difference	95% interval
LR	0.0953	0.0838	0.0114	[-0.0110, 0.0243]
CART	0.0972	0.0784	0.0188	[0.0064, 0.0323]
XGBoost	0.0995	0.0862	0.0133	[0.0004, 0.0256]

Matching test rows reduces the original different-cohort AP gap by 89.7 percent for LR and 86.9 percent for CART. The size of this change establishes test-cohort composition as a material part of protocol comparison. It measures evaluation-population sensitivity; the remaining differences combine training history, training size and, for XGBoost, training-selected configurations, so the reduction is not a causal decomposition of leakage or drift.

4.2 Engineering feature comparisons

Table 3 reports every prespecified feature set. Hazard ratings alone yield AP 0.0596, 0.0738 and 0.0766 for LR, CART and XGBoost. The full-feature gains are 0.0243, 0.0046 and 0.0096, respectively (Table 4). Paired intervals in Table 4 include zero for all three gains. The comparisons quantify the observed contribution of measured signals and its conditional uncertainty.

Table 3. Engineering feature ablations on the common record-order cohort; entries are AP. All three models use the same columns per row.

Feature set	Columns	LR	CART	XGBoost
All features	17	0.0838	0.0784	0.0862
Hazard ratings only	3	0.0596	0.0738	0.0766
Without hazard ratings	14	0.0900	0.0766	0.0856
Geophone only	4	0.0792	0.1000	0.1025
Without geophone	13	0.0734	0.0794	0.0828
Seismic activity only	9	0.0751	0.0761	0.0785
Without seismic activity	8	0.1020	0.0952	0.1292
Shift type only	1	0.0504	0.0824	0.0824
Without shift type	16	0.0837	0.0778	0.0862

Table 4. Full-feature AP minus hazard-ratings-only AP with conditional paired 95 percent intervals.

Model	Incremental AP	95% interval
LR	0.0243	[-0.0039, 0.0619]
CART	0.0046	[-0.0252, 0.0264]
XGBoost	0.0096	[-0.0148, 0.0413]

The geophone-only XGBoost achieves AP 0.1025, while removal of the seismic-activity group gives 0.1292. The latter differs from the full model by 0.0430 with interval [-0.0034, 0.1067]. Together with the LR and CART contrasts, these results show that predictive use of a monitoring group depends on the other available inputs and fitted model. The comparisons hold XGBoost settings fixed within each fold, assessing the sensitivity of that design to information sources. Redundancy, estimation variability and changing phase distributions are candidate explanations for the observed patterns; they are not separately identified here.

4.3 Training prior correction and probability assessment

Training-prior correction improves LR ECE from 0.2904 to 0.0271 and Brier score from 0.1462 to 0.0439, providing a direct adjustment to the probability scale using historical labels. Corrected LR, raw CART and raw XGBoost Brier scores are 0.0439, 0.0433 and 0.0429. Corrected LR Brier skill against a retrospective evaluation-prevalence constant is -0.0751. The constant is an oracle reference used for assessment, with no role in generating forecasts. The correction improves the weighted LR probabilities on this cohort, while the negative skill score records their position relative to that retrospective reference.

Logistic regression: calibration on 2,063 later records

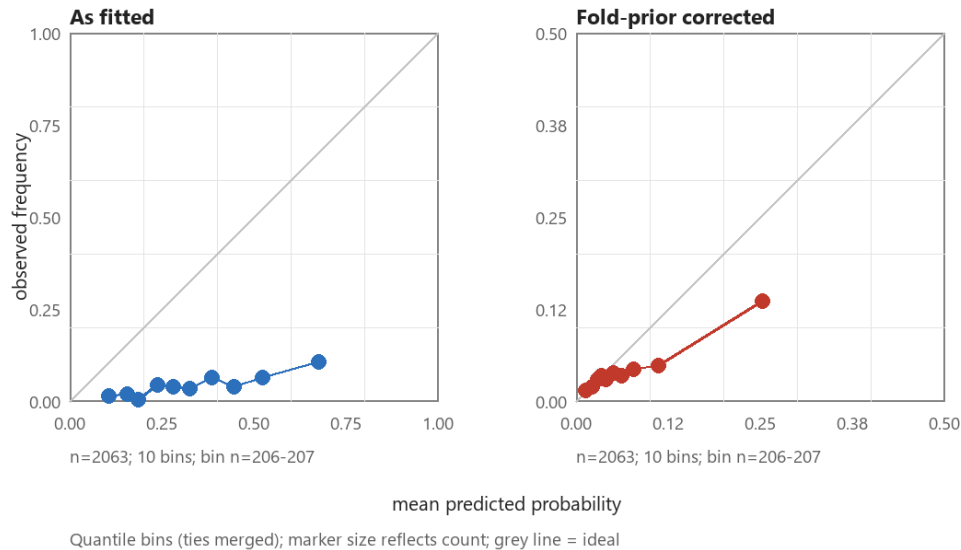


Figure 1. LR calibration on the common test cohort. Each panel contains 10 quantile bins of 206 or 207 observations; ties stay together, marker sizes encode counts, and panel axes differ.

5 Model explanations and phase stability

5.1 Global and stage-specific feature use

Total seismic energy, low-energy event count nbumps2 and pulse count gpuls have the largest mean absolute XGBoost contributions. These are observable indicators of seismic and geophone activity; their importance describes how the fitted models use monitoring information. Table 5 reports the leading contributions and value associations. Read alongside the ablations, these explanations distinguish a model's reliance on a signal from the incremental value of its monitoring group.

Table 5. Leading full-feature XGBoost contributions on 2,063 test rows. Magnitudes are in raw log-odds units; correlations are descriptive value-contribution associations.

Feature	Mean absolute contribution	Value association
energy	0.5642	0.6854
nbumps2	0.3902	0.7179
gpuls	0.3523	0.7668
genergy	0.1466	0.0148
seismic	0.1391	0.8297

XGBoost contributions and stability across four test phases

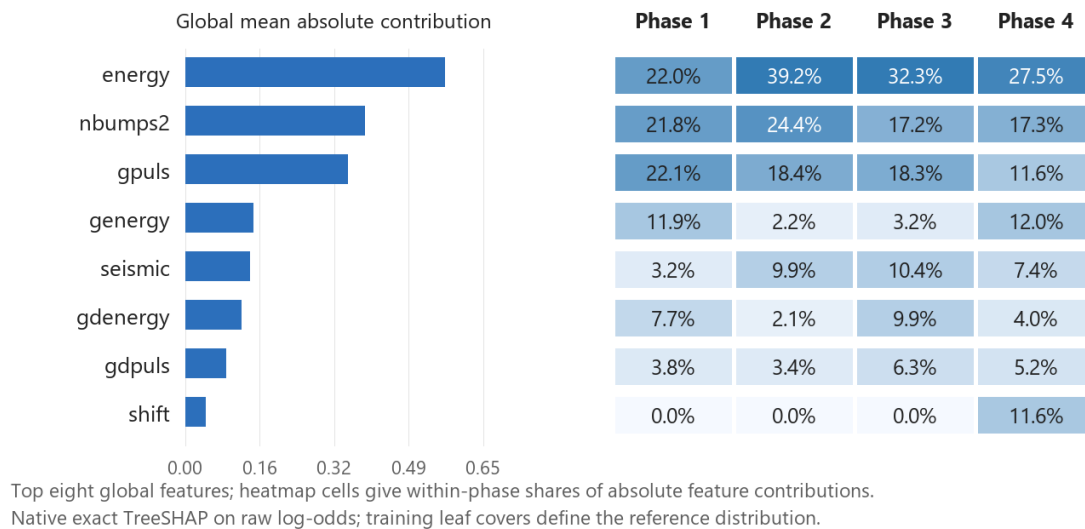


Figure 2. Global importance and phase shares for the eight globally leading features. Each cell is that feature's mean absolute contribution divided by the sum over all 17 features in the phase.

Across the six phase pairs, importance-rank correlations range from 0.7122 to 0.9412 and top-five Jaccard overlap ranges from 0.4286 to 0.6667. This combination shows recurring feature use alongside changing membership of the most influential group. Correlations include all 17 features, including tied low-importance columns; top-five overlap supplies a complementary check focused on leading variables.

Feature values and test-set contributions

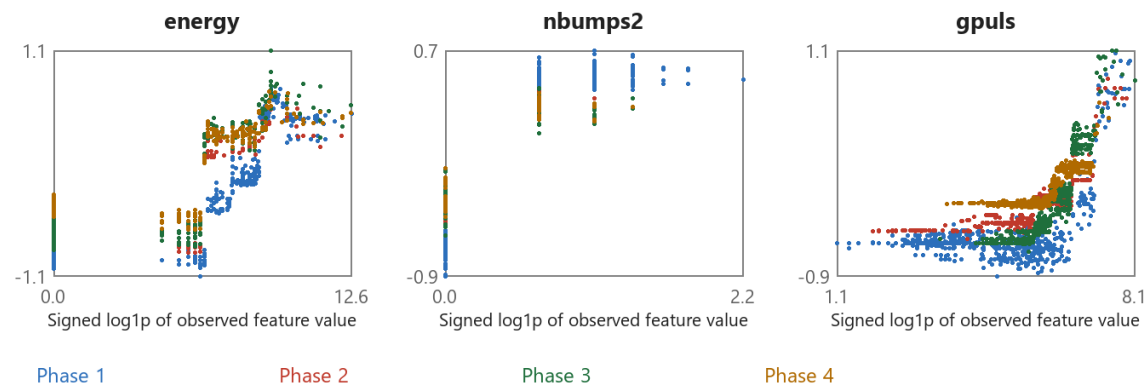


Figure 3. Values and TreeSHAP contributions for the three globally leading features. Colours identify test phases; horizontal values use signed log1p for display, and vertical values are raw log-odds contributions.

The scatter distributions expose phase-dependent and nonlinear feature use. A high global importance can coexist with a weak global monotonic association: genergy, for example, has a descriptive value-contribution correlation of 0.0148. The phase-coloured patterns show how a measurement's contribution varies with the other inputs and the fitted model. For engineering review, the contribution describes the signal in its prediction context.

5.2 Phase outcomes and illustrative warning cases

Table 6 connects explanation changes to the phase outcomes. Positive prevalence and full-model AP vary across the record sequence. Computing AP within each block separates its observed discrimination from the cross-model score-scale differences that can affect pooled AP. Together, Tables 5 and 6 show why monitoring review benefits from examining feature use and phase outcomes side by side.

Table 6. Four common test phases with positive counts, prevalence and within-phase full-feature AP.

Phase	Rows	Positives	Prevalence	LR AP	CART AP	XGBoost AP
1	516	36	0.0698	0.1360	0.1136	0.1341
2	515	10	0.0194	0.0204	0.0274	0.0284
3	516	24	0.0465	0.0873	0.0632	0.0731
4	516	18	0.0349	0.1142	0.0781	0.0618

The predetermined holdout cases are shown in Figure 4. Each case decomposes the bias and all feature contributions to reconstruct the model margin. The five largest absolute contributions are displayed individually, with remaining contributions combined as Other features. The three outcomes show how combinations of measured signals place different records above or below the same frozen cutoff. Signed contributions locate each model adjustment relative to its baseline.

Holdout warning cases at the 10% training reference budget

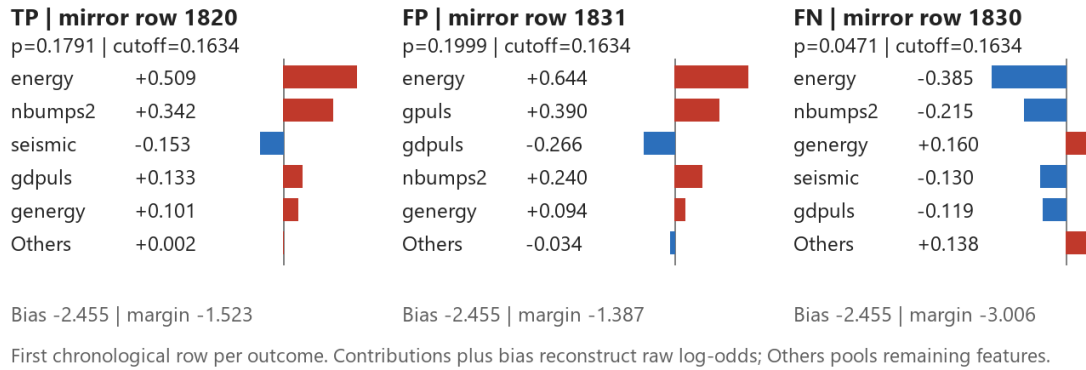


Figure 4. Earliest holdout true-positive, false-positive and false-negative cases under the 10 percent reference budget. IDs are one-based mirror rows; raw log-odds contributions, including Other features, sum with the bias to the margin.

6 Warning budgets and inspection workload

Changing the reference budget changes the frozen threshold while leaving fitted scores unchanged. In the common record-order cohort, LR, CART and XGBoost actual alert rates at the 10 percent reference budget are 0.1609, 0.0785 and 0.0999. This provides a direct comparison between intended historical workload and its transferred operating outcome.

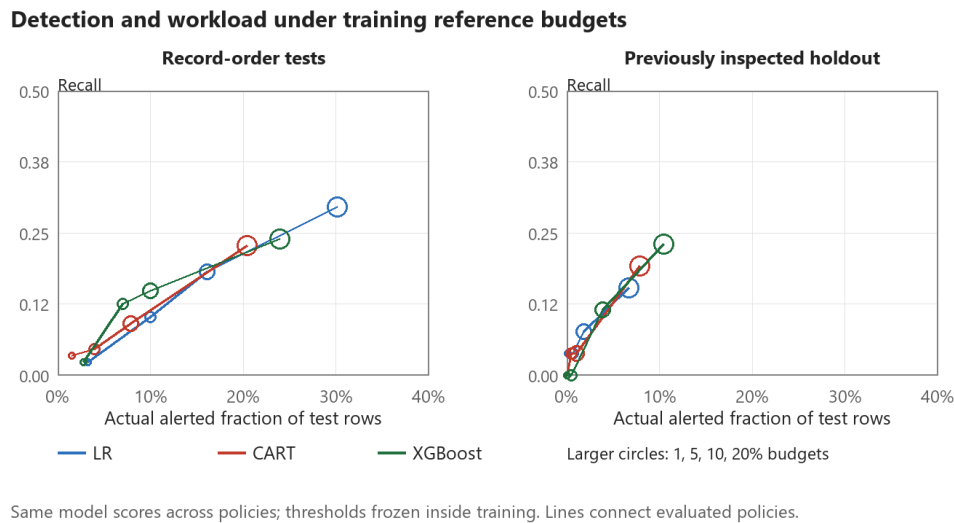


Figure 5. Recall versus actual test alert fraction under the four fixed training reference budgets. Colours identify models and increasing circle sizes identify 1, 5, 10 and 20 percent budgets.

Table 7 reports the supplementary holdout outcomes for every model and budget. For XGBoost, increasing the reference budget from 10 to 20 percent increases alerts from 30 to 81 and detections from 3 to 6, while missed hazardous shifts change from 23 to 20. The corresponding actual alert rates are 0.0388 and 0.1047. Reporting both detection counts and actual alert rates makes the workload change measurable when a historical threshold is transferred to later records.

Table 7. All holdout warning policies on 774 records with 26 positive targets. Budget is the training reference fraction; alert rate is measured on holdout rows.

Model	Budget	Alerts	Detected	False alerts	Missed	Alert rate	Recall
LR	1%	1	1	0	25	0.0013	0.0385
LR	5%	5	1	4	25	0.0065	0.0385
LR	10%	14	2	12	24	0.0181	0.0769
LR	20%	52	4	48	22	0.0672	0.1538
CART	1%	0	0	0	26	0.0000	0.0000
CART	5%	4	1	3	25	0.0052	0.0385
CART	10%	8	1	7	25	0.0103	0.0385
CART	20%	61	5	56	21	0.0788	0.1923
XGBoost	1%	0	0	0	26	0.0000	0.0000
XGBoost	5%	4	0	4	26	0.0052	0.0000
XGBoost	10%	30	3	27	23	0.0388	0.1154
XGBoost	20%	81	6	75	20	0.1047	0.2308

At the 1 percent reference budget, holdout LR issues one alert and detects one hazardous shift, while CART and XGBoost issue no alerts. Detection and missed-shift counts make these sparse outcomes interpretable alongside precision. Higher budgets increase detections together with additional false-alert workload. The resulting policy curves provide evidence for selecting a budget once inspection capacity and the consequences of missed hazards are specified.

7 Discussion

7.1 Connecting monitoring evidence to warning decisions

The engineering contribution is the connection between information sources, model explanations and warning workload. Recorded hazard ratings establish an assessment baseline, while geophone and seismic-activity comparisons measure what monitoring signals add within the tested designs. TreeSHAP then identifies recurring energy, event-count and pulse contributions and shows where their relative importance changes. A high contribution and an incremental AP gain answer different questions: the first describes model reliance, and the second measures the value of adding information to an existing predictor.

Training-prior correction and frozen budgets complete this connection. Probability assessment establishes the effect of restoring the historical prior, and policy evaluation converts fitted scores into detected hazardous shifts, missed shifts and false-alert workload. For an engineer reviewing monitoring forecasts, these outputs support tracing an unusual prediction to its input signals and comparing warning rules against available inspection capacity.

7.2 Scope and prospective validation

These results concern the audited mirror under a record-order forecasting assumption. Timestamps, longwall identifiers and event locations are needed for direct temporal, site-specific and spatial validation. Repeated measurements can correspond to distinct shifts; the provenance audit establishes the mirror transformation rather than the operational validity of deduplication. Native TreeSHAP uses training leaf covers, and correlated inputs affect the allocation of contributions. Its signed values explain fitted associations rather than the causal effect of changing a monitoring variable. Phase stability is conditional on the fitted models and observed test distributions.

The feature comparisons are prespecified descriptive contrasts with unadjusted fixed-prediction intervals. A maximum AP among these feature sets is not treated as a validated model-selection result. The already inspected holdout supplies additional warning evidence but does not constitute an untouched confirmatory test. A subsequent study should preregister its feature and alert-policy choices, then test them on independent timestamped working-face data with defined inspection actions.

8 Conclusions

This study establishes a reproducible connection from mine-monitoring information to explainable seismic warning decisions. Matched test rows make validation gaps interpretable on a common population, and training-prior correction improves the LR probability scale using historical information. Engineering ablations quantify the contribution of monitoring groups, while TreeSHAP identifies recurring energy, event-count and pulse signals and traces changes in their use across phases. Frozen-budget evaluation translates forecasts into detection counts, missed hazardous shifts and inspection workload. The combined evidence provides a practical basis for reviewing which signals a forecasting model uses and what its warning policy requires.

Data and computational reproducibility

Source data are available from UCI [1] and the CSV mirror [2]. Saved evidence includes all feature-set predictions, inner-reference scores, fold audits, candidate-selection predictions and row-index manifests, bootstrap replicates, five random seeds, native XGBoost models, row-level TreeSHAP contributions and the predetermined case records. Each reported table is generated from these files. Source hashes and duplicate row mappings preserve the data version.

The extended analysis uses Python 3.12.14, NumPy 2.3.5, pandas 3.0.1, XGBoost 3.1.3 and SciPy 1.18.1. The legacy LR/CART baselines use the original verified Python 3.7 environment and retain their saved full-model predictions. To reproduce the extensions, install requirements-research.txt in a separate Python 3.12 environment, run `run_engineering.py` followed by `run_research.py`, and execute `verify_research.py --replay`. Report generation uses `generate_report.py --docx`; `export_report.ps1` exports the Word document to PDF. The original experiment and report consistency checks remain available through `verify_results.py --reports`.

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